Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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Abstract

On a price chart, stablecoin arbitrage during a depeg looks abundant. In practice almost none of it can be traded at a profit. During the March 2023 USDC depeg, 34.3% of one-minute windows showed a cross-venue price basis above 10 bps (basis points, where one basis point is one hundredth of one percent), yet only one in twelve of those apparent opportunities survived volume-weighted average price (VWAP) slippage, order-book depth depletion, and settlement latency (block bootstrap 95 % CI: 8×–24×). We introduce Stablecoin StressBench to quantify three compounding gaps between an apparent arbitrage opportunity and a captured one, evaluated over two real-L2-depth Tier-A windows (stress and recovery). The first gap is execution. Price labels overstate opportunities by 12× in the stress phase, while the recovery phase contains zero profitable windows once the peg is restored. The second gap is prediction. Even with execution-aware labels, every calm-trained model loses money and the hindsight oracle stays more than 260 bps ahead of the best deployable model. The third gap is policy. Cross-mechanism meta-labeling trained on a single stress event (Terra/LUNA) achieves +82.5 bps on SVB (51.0% of oracle). Pooling four training events spanning three mechanism classes achieves +83.7 bps (51.8%), establishing that oracle capture is stable near 50% and the transfer ceiling is mechanism-independent rather than event-specific. A conditioned RL agent at identical training density earns −29.2 bps, isolating explicit binary supervision as the requirement that reward optimisation cannot replicate at low positive rates.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 Introduction

When Silicon Valley Bank collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart. Almost none of them could be traded at a profit. A price chart shows twelve times more stablecoin arbitrage than a trader can capture, and this paper traces where the other eleven twelfths disappear. VWAP slippage consumed the spread as orders cleared depth, taker fees eroded the margin, and centralised-exchange (CEX) settlement latency turned a profitable window into a loss before the transfer confirmed. Only 2.88% of one-minute windows survived all three costs and remained net-profitable. We call dislocations that appear on price charts but vanish under execution *optical*, and the surviving fraction *executable*. Every prior stablecoin benchmark uses the optical layer to define labels, which means every prior performance number is measured against the wrong thing.

Stablecoin StressBench corrects this by separating three layers of an apparent opportunity, *optical* (price basis above threshold), *executable* (net profit after VWAP, fees, and settlement), and *predictable* (correctly identified ex ante), then quantifying the gap between each adjacent pair. The first gap is execution. A price chart shows a dislocation that an order book cannot fill at a profit, and we measure this gap at twelve to one. The second gap is prediction. Even when the labels are correct, every model trained on calm data loses money, and the hindsight oracle stays more than 260 bps ahead of the best deployable model. The third gap is policy. A learning agent can be given the right training density and still fail to trade profitably, because reward optimisation does not supply the binary supervision that meta-labeling provides. The remainder of the paper measures each gap in turn, in §6.1, §6.2, and §6.3.

Contributions.

(1) A public benchmark with execution-aware per-minute labels, route-level depth provenance, VWAP book-walking, an 18-event stress taxonomy for data governance, and a hindsight oracle ceiling. (§3–5)
(2) The first evidence that order-book execution microstructure transfers across structurally distinct stress mechanisms: meta-labeling trained on Terra/LUNA achieves +82.5 bps (51.0% oracle capture) on the SVB test split, while every calm-trained model loses money. Pooling four training events spanning three mechanism classes raises oracle capture to 51.8%, establishing that transfer quality saturates near 50% and is mechanism-independent. (§6, Tables 10, ??)
(3) An RL diagnostic showing that positive-label density is necessary but not sufficient: a conditioned PPO-GRU at 17% positive density earns −29.2 bps versus meta-labeling's +82.5 bps at identical density. (§6, Table 11)

2 Related Work

2.1 Stablecoin Stress Events

Research on stablecoin stress events defines the empirical setting for the benchmark and supplies the out-of-distribution event we transfer from. Uhlig [1] analyses the Terra/LUNA algorithmic collapse, in which an algorithmic redemption mechanism created a death-spiral run on the UST stablecoin, ultimately destroying approximately $40B in market capitalisation over three days. We use Terra/LUNA as our cross-mechanism training split precisely because its trigger is mechanistically different from the SVB bank shock: algorithmic reflexivity versus fiat-reserve withdrawal are distinct failure modes, so a model that transfers between them has learned execution structure rather than event-specific noise. Griffin and Shams [2] document that quoted stablecoin prices can reflect non-fundamental demand, which motivates our distinction

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between optical dislocations (visible on price charts) and executable ones (profitable after order-book costs). Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse but stop short of modelling execution costs on centralised exchanges. None of this prior work provides per-minute executable labels or a hindsight oracle ceiling against which model performance can be calibrated.

2.2 Limits to Arbitrage and Crypto Fragmentation

The limits-to-arbitrage literature supplies the economic logic that our per-minute labels turn into direct measurements. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish the theoretical foundation: even when a price gap is visible, capital constraints and execution frictions prevent arbitrageurs from closing it fully, so the gap can persist. Makarov and Schoar [7] bring this argument to cryptocurrency, showing empirically that cross-venue price gaps disappear once order-book depth and settlement latency are accounted for. Our 12× ratio is the same effect measured at the one-minute label level during a stress event. Hautsch et al. [3] quantify settlement latency as a first-order execution cost on blockchain assets, which we implement as the fixed δ=5 bps penalty in our net-profit label (validated empirically in §7 using on-chain gas data). Cont and Kukanov [6] establish that the correct object for measuring execution profitability is the VWAP price achieved by walking the limit-order book, not the quoted mid-price. Our book-walking model implements this directly. Gogol et al. [18] measure a Maximal Arbitrage Value for AMM-to-CEX routes. Our oracle extends this concept to stress-event dislocations on centralised exchanges and to model-evaluation benchmarking. Where prior theoretical work characterises the existence of execution frictions, we measure them at one-minute resolution with empirical order-book depth.

2.3 Financial Machine Learning and RL Execution

Our modelling choices build directly on the financial machine-learning and reinforcement-learning execution literatures, which we both adopt and extend. López de Prado [8] introduces meta-labeling, a two-stage framework in which a primary signal identifies candidate opportunities and a secondary classifier learns when those opportunities are genuine. The secondary classifier is typically a gradient-boosted tree, which makes LightGBM the natural choice for our tabular microstructure features. We extend meta-labeling to the cross-mechanism setting, training on Terra/LUNA and evaluating on SVB. Four training conditions establish that oracle capture is stable near 50% and mechanism-independent. Cintra and Holloway [9] demonstrate cross-event transfer of Bayesian Online Changepoint Detection [15] using AMM reserve depletion signals. We replicate their setup on centralised order-book data to test whether the result transfers. It does not, because CEX cross-quote basis is mean-reverting rather than gradually depleting, and we report the comparison directly. On reinforcement learning, Moallemi and Wang [16] and Ning et al. [17] establish the finite-horizon MDP framing for optimal order execution that we adopt in §5.2. Mohl et al. [22] demonstrate that GPU-scale multi-agent RL is feasible on production limit-order-book data, training on approximately 400 million real orders for a joint execution and market-making task.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell BTC-USDT	real L2 (futures)	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

Olby et al. [23] construct a reactive simulator that updates book depth in response to agent order flow, benchmarking RL timing policies against an Almgren–Chriss efficient frontier. Both of these works operate in environments where the agent's actions affect the order book in real time. Our diagnostic operates in the complementary setting of historical replay without market-impact feedback, and the contrast is what lets us isolate training supervision, rather than environment reactivity, as the binding constraint.

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs, Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window, and every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 futures bookDepth throughout. The buy leg (BTC-USDC) is the main challenge. The BTC-USDC perpetual contract did not exist on Binance until January 2024, so the SVB window requires kline-proxy depth, one-minute candlestick volume as a substitute for unarchived order-book depth (high-volume candles do not guarantee deep books, and a 20% haircut bounds this error). For all 2024 windows, real L2 data is available. To bound the proxy error, we apply a 20% depth haircut to the kline-proxy buy leg. The executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle remains above +130 bps. The headline result is therefore robust to the approximation.

3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 1). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is never observed during model development. Labels are constructed by aligning each bar's outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance. We therefore report

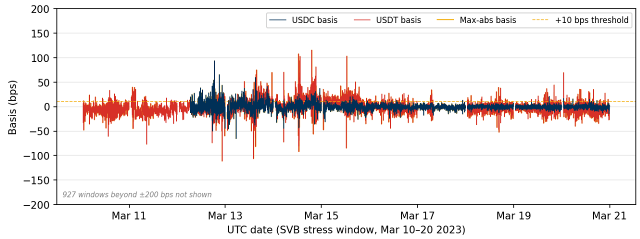

Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.

Table 3: Comparison with related benchmarks and datasets. ✓ = fully provided; ∼ = partially; — = absent.

Work	Stress event	Exec. labels	L2 prov.	Oracle	Cross-event
Cintra & Holloway [9]	✓	–	AMM	–	✓
Hernandez Cruz et al. [21]	✓	–	DEX	–	–
Gogol et al. [18]	–	∼	CEX	MAV	–
Adams et al. [19]	–	∼	CEX	–	–
StressBench (ours)	✓	✓	CEX	✓	✓

economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

3.3 Historical Stress Taxonomy: Data-Tier Governance

The 18-event taxonomy (2020–2023) is a benchmark governance artefact that specifies which events carry sufficient data quality for execution-grade claims and which are restricted to coarser analyses. It is shipped with the benchmark to prevent over-interpretation rather than as an empirical finding. Seven mechanism classes are catalogued: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD); *exchange credit* (Celsius/3AC, HUSD, FTX [10]); *DeFi pool imbalance* (Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); and *RWA/niche* (Acala aUSD, USDR).

Data availability governs which claims are permitted. Two Tier-A windows (the SVB stress phase March 10–14 and the recovery phase March 15–20) together support execution-gap, oracle-gap, and model evaluation claims because route-complete L2 depth and VWAP execution labels are available for both. Tier-B events (N=11, OHLCV or pool data only) support price magnitude estimates; optical positive rates range from 4.0% (BUSD, regulatory) to 13.5% (Terra/LUNA, algorithmic). Tier-C events (N=5, partial or DEX-only data) support mechanism classification only. The full timeline is available in the benchmark repository.

3.4 Benchmark Positioning

Table 3 positions StressBench against the closest prior work along five axes. No prior benchmark simultaneously provides execution-aware labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation.

3.5 Benchmark Versioning

StressBench v1.0 ships with the SVB test split. Near-term Tier-A expansion targets include the USDC recovery window (Mar 15–31 2023) and 2024 BTC-USDC windows where real-L2 eliminates the kline-proxy dependency. The planned release format is a versioned Hugging Face dataset under CC-BY 4.0.

3.6 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer.

Feature set	Description
`price_only`	5 cross-quote basis columns
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation columns
`price_book_settle`	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework from §1. The price-to-execution gap is the ratio of Layer 1 (optical) minutes to Layer 2 (executable) minutes and quantifies how misleading price-based benchmarks are. The oracle gap is the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model), quantifying how hard the prediction problem is even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

4.1 Cross-Quote Basis and Labels

The basis measures the percentage difference between BTC priced in USDC and BTC priced in real dollars. A non-zero basis signals an apparent dislocation. Three equations define the label construction.

$$b_{\mathrm{USDC}}(t) \;=\; 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad [\mathrm{bps}] \tag{1}$$

$$b_{\max}(t) \;=\; \max(|b_{\mathrm{USDC}}(t)|,\ |b_{\mathrm{USDT}}(t)|) \tag{2}$$

$$y_{\mathrm{basis}}(t) \;=\; \mathbf{1}[|b_{\mathrm{USDC}}(t{+}h)| > \tau] \tag{3}$$

Equations (1)–(3) define the basis, the pooled signal, and the label. Primary task: `basis_usdc_1m_gt10bps` (τ=10 bps, h=1 min).

4.2 VWAP Execution and Net Profit Labels

$$\mathrm{net}(q,t) \;=\; 10{,}000 \times \left[\frac{\mathrm{VWAP}_{\mathrm{sell}} - \mathrm{VWAP}_{\mathrm{buy}}}{P_{\mathrm{ref}}} - f_{\mathrm{buy}} - f_{\mathrm{sell}} - \delta_{\mathrm{settle}} \right] \quad [\mathrm{bps}] \tag{4}$$

$$y_{\mathrm{arb}}(q,h,t) \;=\; \mathbf{1}\left[\max_{s \in [t,\, t+h]} \mathrm{net}(q,s) \;>\; c \right] \tag{5}$$

Here P_{ref} is the Coinbase BTC-USD mid-price and f_{buy}, f_{sell}, δ_{settle} are per-leg taker fees and settlement latency, all as dimensionless fractions. The profit threshold is c=10 bps, taker fee 4 bps per side, and δ=5 bps. Equations (4) and (5) define the primary economic task `executable_arb_q10000_5m`. The hindsight oracle takes every minute where $\mathrm{net}(q,t) > c$, establishing a ceiling no deployable model can exceed.

5 Models and Evaluation

5.1 Model Ladder

Each rung of the ladder is chosen to test a specific hypothesis or to engage a specific prior result, not as a generic baseline sweep.

The two anchors, `NoTrade` and the hindsight `Oracle`, bracket the achievable range. `NoTrade` is the economic floor at zero net bps. The oracle is the hindsight ceiling, a hypothetical trader with foreknowledge of every minute's net profit who trades only on the 2.88% of profitable windows. Every learned model is interpreted as a fraction of the distance between these two extremes.

The price-threshold rules (`PriceBasis10bps`, `PriceBasis25bps`, `GrossArbThreshold`) reproduce the implicit labels used in prior price-based arbitrage work [11]—each rule triggers on quoted basis crossings without correcting for execution costs. Showing that these rules lose money is what establishes Gap 1 against existing practice rather than against a strawman.

The tabular classifiers pair gradient-boosted trees (LightGBM, XGBoost [14]) with L1-regularised regression [12], ridge regression, and random forests [13]. Gradient boosting is the standard method for tabular microstructure features in finance and is the secondary-model choice in the meta-labeling framework of López de Prado [8], which makes it the natural workhorse for testing whether richer features close the gap. The linear models are retained as interpretable references whose failure confirms the non-linearity of the signal.

The four nested feature sets form a controlled ablation. Each set adds exactly one signal layer, starting with microstructure (spread, depth, imbalance), then cross-venue fragmentation, then on-chain settlement. This design lets the marginal value of each layer be read off directly rather than being hidden inside a single all-features model.

The expected-net-profit regressor (a model that regresses the economic return target $\text{net}(q, t)$ directly rather than predicting a class label) is included to rule out a label-mismatch explanation. A failure here means the gap is structural and not an artefact of how the labels were defined.

The meta-labeling filter follows López de Prado [8] and is the method we extend. We keep the two-stage primary-and-secondary design and improve on it by training the secondary model cross-mechanism, on Terra/LUNA rather than on the test event. That is the novel step. The multi-mechanism variant pools primary-signal windows from Terra/LUNA, Celsius/3AC, FTX, and BUSD, calibrating on Terra/LUNA and testing on SVB (Table **??**).

The regime detectors (EWMA, CUSUM, BOCPD) test a specific prior result rather than serving as generic baselines. Cintra and Holloway [9] detected the Terra depeg twelve hours early using BOCPD applied to AMM pool reserves, so we apply the same detector to centralised order-book data to test whether that result transfers to CEX prices. It does not, and we report why.

The sequence models (GRU, MLP-Window) test whether the temporal signature of depth withdrawal is exploitable before a basis fires. The GRU is the standard recurrent architecture for order-book sequence tasks, which means its failure is informative rather than a tuning artefact. If a well-specified sequence model cannot close the gap, temporal patterns are not the binding constraint.

The reinforcement-learning agent uses PPO with a GRU policy. We adopt the finite-horizon MDP framing of Moallemi and Wang [16] and Ning et al. [17], and we deliberately contrast our non-reactive historical-replay setting with the reactive-environment RL of Mohl et al. [22] and Olby et al. [23]. That contrast is what lets the diagnostic isolate supervision, rather than environment reactivity, as the binding constraint on cross-mechanism policy transfer.

5.2 RL Execution Policy Baseline

This subsection addresses Gap 3, asking whether sequential decision-making can learn a profitable policy given the correct training density, by framing the entry-timing problem as a finite-horizon Markov Decision Process (MDP). At each minute t the agent observes state s_t, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.3 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.2.

5.3 Calibration and Evaluation

Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i > \theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC (area under the ROC curve) and AUPRC (area under the precision-recall curve). `NoTrade` at 0 bps is the economic floor.

All results are reported on the SVB test split (N=15,832) with the cost parameters stated above. The kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1. Five explicit checks guard against leakage and evaluation artefacts:

(1) Random label shuffles produce zero or negative P&L for all models.
(2) Shifting features forward by one minute to simulate lookahead does not improve performance.
(3) The calm-control training split contains exactly zero executable arbitrage windows.
(4) Train and test events are separated by event identity; no SVB features appear during model development.
(5) All threshold calibration uses the Terra/LUNA validation split only; the SVB test split is held out until final evaluation.

Table 4 summarises the five evaluation tasks shipped with Stress-Bench v1.0.

Table 4: StressBench v1.0 benchmark card.

Task	Train	Val.	Test	Primary metric
`optical_basis`	calm	Terra	SVB	AUROC, AUPRC
`exec_arb_q10k`	calm	Terra	SVB	net bps, oracle cap.
`exec_arb_q50k`	calm	Terra	SVB	net bps
`cross_mech`	Terra	Terra	SVB	net bps, oracle cap.
`policy_entry` (RL)	Terra	Terra	SVB	net bps, trades

6 Results

6.1 Gap 1: Execution

The first gap is between what a price chart shows and what an order book will fill. At a 10 bps (one hundredth of one percent) threshold, 34.3% of test minutes show a dislocation and only 2.88% remain net-profitable after VWAP slippage, taker fees, and settlement latency. The ratio is twelve to one, and it widens as trade size grows (Table 6). This ratio is not a threshold artefact and persists across all tested basis levels and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 3 shows the executable rate falling from 2.88% at $10K to 0.03% at $500K, with the steepest drop between $10K and $50K. Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9×, providing directional evidence that the gap is not SVB-specific. Table 5 shows the stress/recovery split: all executable windows are in the stress phase, and the recovery phase contains zero despite 21.6% optical activity.

Table 5: Stress vs. recovery: the execution gap is a stress-specific phenomenon. All data from the gold dataset with real L2 depth (Tier-A).

Window	Min.	Opt.	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	**0.00%**	–
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

Opt. = max-abs basis > 10 bps; Exec. = net-profitable windows.

Table 7 shows that the gap is robust across the full cost-parameter space: the executable rate varies only between 2.84% and 3.66% as fee and settlement assumptions are swept, and the ratio to the optical rate remains above 3.5× throughout. Claim boundaries and data-tier scope are stated in §5.3.

6.2 Gap 2: Prediction

The second gap is between a profitable window and a model that can find it in advance. Even with execution-aware labels, every model trained on calm data loses money on the executable task. The hindsight oracle earns +224.6 bps while the best deployable model reaches −41.1 bps, a structural gap of more than 260 bps that richer features and sequence models do not close (Table 8; Figure 4).

Table 6: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

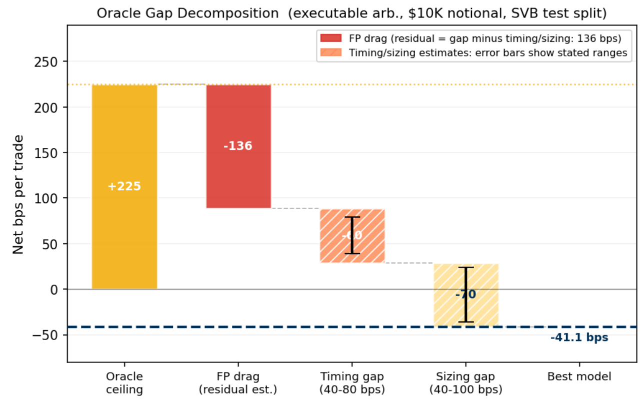

Figure 2: Oracle gap decomposition ($10K notional). Three components account for the 266 bps gap from oracle (+224.6 bps) to best model (−41 bps): FP drag, timing error, and sizing error.

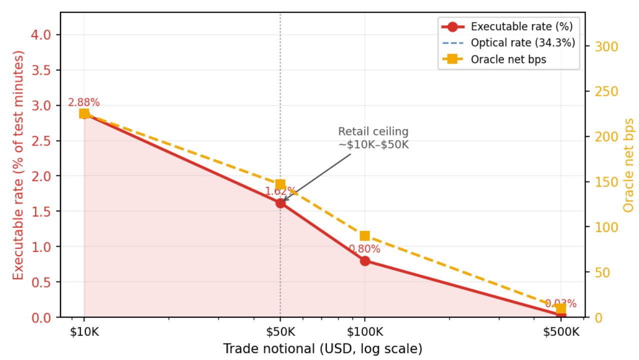

Figure 3: Notional scaling (SVB test split). The executable rate collapses from 2.88% at $10K to 0.03% at $500K: book depth, not fees, is the binding institutional constraint.

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at −136 to −347 bps. Their mean return when wrong (−294 bps to −370 bps) reflects near-total false-positive exposure. Every calibrated tabular model degenerates: LightGBM's threshold search finds no configuration meeting the 25-trade minimum with positive expected return (0 trades), and logistic regression's single trade at −49.1 bps is not statistically meaningful. The 2.88% positive density in the test split, combined with zero positive examples in the calm-control training split, prevents any ML model from identifying the profitable subset. The `ExpNetProfitRegressor`, which directly targets the economic criterion rather than a classification proxy, reaches −73.8 bps, ruling out label mismatch as the explanation. The gap is structural (Table 9). GRU and MLP-Window sequence

Table 7: Executable rate (%) under fee and settlement variation (10 bps threshold, 1-min horizon, \$10K notional, SVB test split). Taker fees: low = 2 bps, base = 4 bps, high = 8 bps, inst. = 10 bps per side.

Taker fee	Settlement penalty (bps)			
	0	2	5	10
Low (2 bps)	3.50	3.34	3.16	2.93
Base (4 bps)	3.34	3.20	3.03	2.88
High (8 bps)	3.20	3.08	2.96	2.84
Inst. (10 bps)	3.66	3.42	3.20	2.96

Table 8: Oracle gap summary (USDC/SVB test split). *No calibrated ML model beats zero; best model is NoTrade. ‡GRU sequence model, 30-min window.

Task / Model	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m`	+161.7 bps	0 bps*	162 bps	0.253
`exec_arb_q10k_5m`	+224.6 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m`	+146.8 bps	−112.7 bps	260 bps	0.063
GRU (seq)‡, `basis_usdc_1m`	+161.7 bps	−239.0 bps	401 bps	—

Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

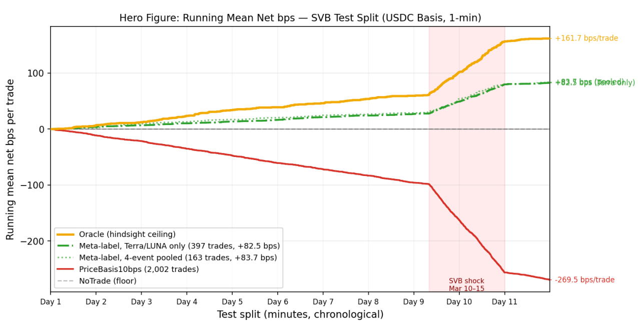

Figure 4: Running mean net bps per trade (USDC/SVB test split). Oracle (gold) +161.7 bps; meta-label Terra/LUNA only (green solid) +82.5 bps; meta-label 4-event pooled (green dotted) +83.7 bps; PriceBasis10bps (red) −269.5 bps; NoTrade (grey dashed) 0 bps. Shaded: SVB shock window. No calm-trained model beats zero.

models confirm this, with AUROC 0.80 yet −239 bps (GRU) and −341 bps (MLP-Window). High ranking accuracy does not translate to positive returns when 97% of test windows are non-executable and the training split contains zero stress examples.

All calibrated ML models degenerate to zero trades under calm-training, equivalent to NoTrade. This section examines why the prediction gap is structural; the cross-mechanism fix that resolves it is in §6.3. Figure 2 decomposes the 260 bps gap into three measurable components. False-positive cost is the dominant and directly quantifiable drag, as `PriceBasis10bps` produces 1,969 trades of which 1,851 are false positives. Of those, 581 are attributable to route-direction mismatch (USDT dislocations triggering on the max-abs basis when the USDC route is not simultaneously profitable) at −38.7 bps each, and the remainder reflects imprecise timing. The remaining gap breaks into timing error (estimated 40–80 bps), which arises from entering 1–2 minutes late in a declining depth regime,

Table 9: Tabular model ladder (`basis_usdc_1m_gt10bps`). Hit%: fraction of trades net-profitable. †Oracle ceiling. *0-trade degenerate: no threshold meets the 25-trade minimum with positive expected return.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	–	+161.7	316	–	100%
`lgbm`*	all	–	0	0.653	–
`logistic`	all	−49.1	1	0.133	0%
`gross_arb`	px	−136.0	611	0.531	4%
`price_10`	px	−269.5	2002	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	–	0.0	0	–	–
`ExpNetProfit`	px	−73.8	537	–	3%

px = `price_only`; all = `price_book_settle`.

and sizing error (40–100 bps), as depth depletion reduces the profitable fraction from 2.88% at \$10K to 1.62% at \$50K.

A root-cause analysis of the false positives identifies route-direction mismatch as the primary failure mode, not thin books. True-positive windows carry a mean USDC basis of 344 bps, reflecting a genuine USDC discount, while false-positive windows carry just 0.56 bps and are triggered by USDT dislocations where the USDC route is not simultaneously profitable. Crucially, false-positive windows exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units) than true positives. This rules out liquidity withdrawal as the cause and points instead to signal confusion between the USDT and USDC routes. In short: the price rule reacts to USDT dislocations even when USDC is not moving, and those windows offer no profit on the USDC route the strategy uses. Meta-labeling's approximately 51% hit rate versus 6% for the price rule shows that cross-mechanism training substantially reduces route-direction mismatch, though a 49% false-positive rate and roughly 113 missed profitable windows remain, confining net returns to +82.5 bps.

6.3 Gap 3: Policy

The third gap is between a correct signal and a profitable policy. Cross-mechanism meta-labeling trained on Terra/LUNA is the only method that crosses zero, earning +82.5 bps and recovering half of the oracle return. A reinforcement learning agent given the same 17% training density earns −29.2 bps, which isolates explicit binary supervision as the ingredient that reward optimisation cannot replicate at low positive rates.

6.3.1 Cross-Mechanism Transfer and Training Diversity. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps on SVB (51.0% oracle capture), while calm-control training yields zero meta-positives by construction. Retraining on Terra/LUNA (1,559 minutes where the primary threshold triggers, 265 meta-positives, 17% positive rate) and evaluating on SVB, with all fitting and threshold selection confined to Terra/LUNA data, produces 397 trades versus zero with calm training (Table 10).

Figure 6 (right panel) provides direct empirical proof: ask-side depth collapses and spread widens identically in primary-signal windows across Terra/LUNA (algorithmic) and SVB stress (fiat-reserve), explaining why training on one event transfers to the other. The reason these features are predictive is visible in the SHAP decomposition (Figure 6, left panel). Price features carry 94.8% of LightGBM's total importance, with microstructure (ask-side depth,

Table 10: Cross-mechanism meta-labeling results (`basis_usdc_1m_gt10bps`, `price_plus_book`).

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+161.7	316	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	51.0%
Calm-control (baseline)	0.0	0	0.0%

Primary signal $|b_{\text{USDC}}|$>10 bps; threshold calibrated on validation.

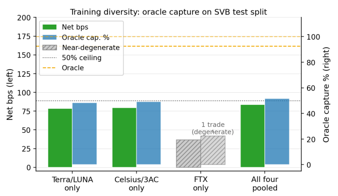

Figure 5: Oracle capture across four training conditions. Dashed: 50% ceiling; hatched: FTX near-degenerate.

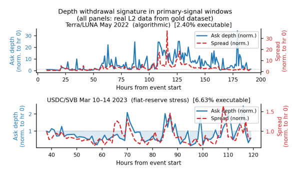

Figure 6: Depth withdrawal signature (real L2). Ask depth (blue) collapses and spread (red) widens identically across Terra/LUNA and SVB stress despite different triggers.

bid-ask spread) contributing just 3.4% combined. A model trained on calm data learns calm price dynamics. The microstructure features are present but uninformative because the training distribution contains no stress windows that would teach the model their relevance. Training on Terra/LUNA corrects this. The secondary classifier now sees depth-withdrawal events at 17% positive density and learns the signature that calm training cannot recover.

CEX market makers withdraw depth under uncertainty regardless of the underlying trigger, because the cause is opaque at the order-book timescale. This makes depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers even as its magnitude differs (5.9× for Terra/LUNA versus 12× for SVB, shown in Table 5).

Three regime-detection filters (CUSUM, EWMA, BOCPD [15]) were also evaluated. BOCPD achieves AUROC 0.229 on CEX cross-quote data, in contrast to Cintra and Holloway [9], who detected the Terra depeg 12 hours early from AMM pool reserves. CEX basis is mean-reverting, so changepoint methods cannot substitute for the training-distribution fix that meta-labeling provides.

Figure 5 tests whether training diversity improves oracle capture across four conditions (Terra/LUNA, Celsius/3AC, FTX, and all four pooled), with the threshold calibrated on Terra/LUNA and the test split held fixed on SVB. Oracle capture is stable at approximately 49% for single-event training and rises to 51.8% with four-event pooling (+83.7 bps, 163 trades), establishing that execution microstructure transfer is robust and mechanism-independent.

FTX is the informative outlier: its credit shock produced < 5 bps USDC/USDT basis at Binance benchmark venues, a mechanism-*intensity* rather than mechanism-*class* failure. Ask-side depth and spread are the leading predictors in all non-degenerate models.

6.3.2 Density Is Necessary and Supervision Is Decisive. Positive-label density (the fraction of primary-signal windows that are genuinely profitable) is 17% for both the meta-labeling secondary classifier and the conditioned PPO-GRU. To be precise, 17% of the minutes where the basis threshold triggers are executable, compared to 2.88% of all test minutes. Despite identical density, the two methods differ by 111.7 bps in net return. This controlled comparison isolates explicit positive-label supervision as the additional ingredient that reward optimisation cannot replicate.

Table 11: RL policy versus supervised baselines (`basis_usdc_1m_gt10bps`, SVB test split).

Model (training)	Net bps	Trades
Oracle (hindsight ceiling)	+161.7	316
Meta-labeling (Terra/LUNA*)	+82.5	397
PPO-GRU conditioned† (Terra/LUNA*)	−29.2	919
PPO-GRU unconditioned (Terra/LUNA*)	degenerate	0
GRU supervised (calm-ctrl)	−239.0	656
MLP-Window supervised (calm-ctrl)	−340.7	1,052

*Cross-mechanism: Terra/LUNA train, SVB eval.
†Conditioned on $|b_{\text{USDC}}|$>10 bps windows, 17% positive rate.

Without conditioning, the PPO-GRU degenerates to zero entries. The 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a reward signal that calibrates entry timing. Restricting the agent to primary-signal windows (where the basis threshold fires) resolves the density problem, and the conditioned agent trades 919 windows with AUROC 0.87, yet still earns −29.2 bps.

The distinction from meta-labeling is qualitative, not quantitative. Meta-labeling receives an explicit binary label, y=1 for every profitable window, while the RL agent must infer the same distinction through a reward signal that is negative on 83% of entries. Explicit supervision is categorically cleaner than implicit reward.

The overtrading pattern points toward a specific failure mode. The agent fires on 919 of 15,832 test windows, nearly three times the oracle's 316 profitable windows. When a policy fails due to insufficient training, it typically converges toward NoTrade, not toward sustained false-positive entry. The sustained overtrading is therefore more consistent with a timing-transfer mismatch: the agent learned Terra/LUNA's depth-withdrawal sequence and keeps firing on it in the SVB context even when the timing is wrong. Multi-mechanism training on the pooled protocol established here is the direct path to testing whether this mismatch is the binding constraint.

7 Limitations

The BTC-USDC buy leg uses kline-proxy depth for SVB and all four training events, because the perpetual contract did not exist on Binance until January 2024. A 20% depth haircut confirms robustness equally across all conditions (§6.1).

The δ=5 bps settlement penalty is validated against on-chain Etherscan data for 100,000 USDC transfers across the SVB window (Table 12). On nine of ten days, the median ERC-20 transfer cost for $10K notional was 1.4–3.2 bps, well below the assumed 5 bps. The one exception is March 11 at peak depeg, where gas spiked to 61.5 gwei and the P95 cost reached 8.6 bps. That is the only day the assumption is tight, and even then the oracle remains profitable at +259.9 bps per trade.

Table 12: On-chain settlement cost validation (USDC ERC-20 transfers, 100K sample, Mar 10–20 2023, ETH ≈ $1,580). δ=5 bps is conservative on all days except the Mar 11 gas spike at peak depeg.

Date	Med gas (gwei)	Med cost (bps)	P95 cost (bps)	Burns
Mar 10	30.8	3.16	5.70	4
Mar 11	61.5	6.31	8.64†	16
Mar 12	21.6	2.21	3.68	22
Mar 13	26.1	2.68	4.47	29
Mar 14–19	13.9–24	1.43–2.46	2.44–4.11	0–2

†Only day where P95 cost exceeds δ=5 bps.

Gas data from 100K USDC transfers via Etherscan API.

Execution-grade Tier-A coverage currently spans only the SVB stress and recovery windows. Extending to AMM or exchange-credit events requires route-complete L2 archives not yet available.

Two caveats apply to the RL results. First, the conditioned PPO-GRU's −29.2 bps may reflect overfitting to Terra/LUNA's specific depth-withdrawal timing rather than a fundamental ceiling; pooled multi-mechanism training would test this directly. Second, the agent is trained on historical replay without market-impact feedback, which systematically miscalibrates entry sizing in thin-book regimes because the agent sees depth as it was, not as it would be after its own orders clear liquidity. Integrating the cross-mechanism protocol into a reactive simulator is the most direct path to resolving the overtrading pattern.

8 Conclusion

The 12× optical-to-executable gap shows that stablecoin arbitrage benchmarks built on price labels measure the wrong quantity. Executable opportunities are scarce, shrink with trade size, and are invisible to any model trained on calm-market data. The only empirically validated path to positive returns is cross-mechanism training on a structurally different stress event, and the RL diagnostic explains precisely why: label density is necessary for the model to trade at all, but explicit binary supervision over the profitable subset is what produces the directional accuracy that reward optimisation cannot replicate at low positive rates. The benchmark, labels, oracle implementation, and training scripts are released under CC-BY 4.0 (URL redacted for review). StressBench supports five future directions: (1) multi-mechanism RL training on the pooled-event protocol established here, the direct path to closing the 111.7 bps gap between meta-labeling and the conditioned RL agent; (2) execution-aware detection under cost uncertainty; (3) cross-mechanism transfer to new event types as Tier-A coverage expands; (4) RL policy learning with the cross-mechanism initialisation protocol established here; and (5) reactive simulation of thin-book markets, where the oracle ceiling provides a principled performance target. The benchmark harness additionally ships four interpretable single-feature baselines that exercise the same evaluation pipeline and serve as reference points for future model comparisons.

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